Composition Determines Diversity

Categorical Fingerprints of Genetic Algorithms

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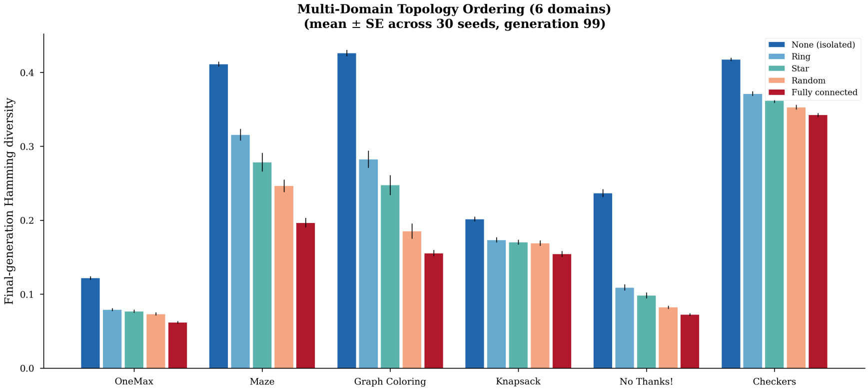
"How you wire your islands matters more than what you evolve."

The Surprising Claim



Kendall's W = 1.0 **$p = 0.00008$** **Topology explains 23.9× more variance than domain**

Why Should You Care?

What practitioners tune

- Operator selection
- Crossover type
- Mutation rate
- Population size

What actually determines diversity

- **Composition structure**
- **Migration topology**

Three recommendations: (1) Prefer ring/mesh over star. (2) Monitor λ_2, not diversity. (3) Close cycles, don't just raise migration rate.

The Problem: Hidden Composition

```
// selection
let idx = tournament_select(
&pop, &cfg, rng);
// crossover
let child = one_point_crossover(
&pop[idx], &mate, rng);
// mutation
let mutant = point_mutate(
&child, &cfg, rng);
// evaluate
let fit = evaluate(&mutant, &env);
pop[i] = (mutant, fit);
```

- Pipeline exists only as **statement order**
- Selection returns an **index**, not an individual
- Randomness & config **explicitly threaded**
- Composition has **no formal representation**

You can't ask: does this

Making Composition Explicit

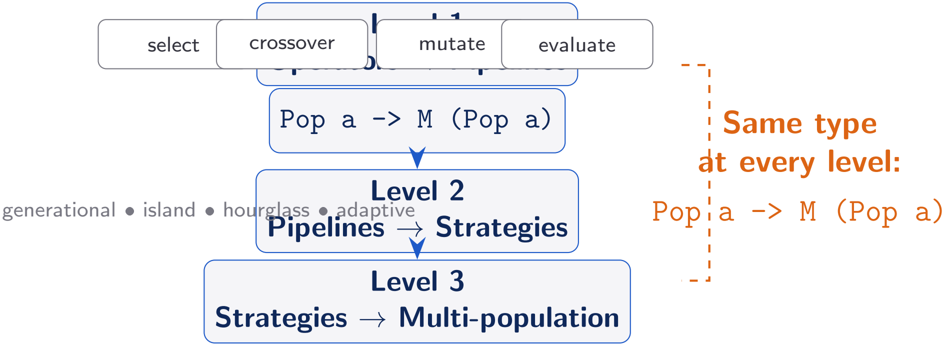
```
let idx = tournament_select(
&pop, &cfg, rng);
let child = one_point_crossover(
&pop[idx], &mate, rng);
let mutant = point_mutate(
&child, &cfg, rng);
let fit = evaluate(&mutant, &env);
pop[i] = (mutant, fit);
... (15+ lines)
```

```
generationStep fitFunc
mutFunc gen =
evaluate fitFunc
>>>:  logGeneration gen
>>>:  elitistSelect
>>>:  onePointCrossover
>>>:  pointMutate mutFunc
```

6 lines. Same algorithm.

>>>: is Kleisli composition — threads randomness, config, and logging automatically. **The composition is now a first-class value you can type-check, compose, and reason about.**

Three Composition Levels

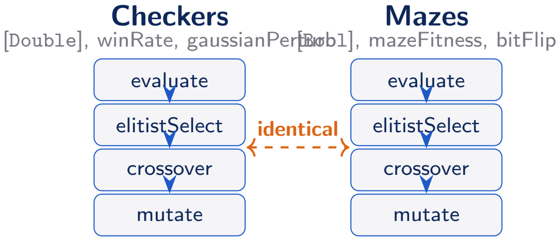


The composite at every level has the same type as its parts — a design algebra, not just notation.

Domain Invariance

Changes	Invariant
Genome type	Pipeline shape
Fitness function	Monad stack
Mutation operator	Composition operator
	Strategy combinators

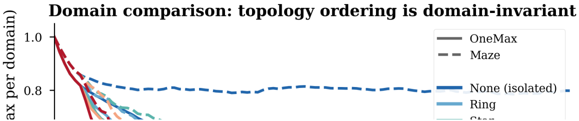
Category theory predicts: structure lives in the arrows, not the objects.



Six Domains: Maximally Diverse

1. **OneMax** — trivially unimodal, binary
2. **Maze** — multi-objective, binary
3. **Graph Coloring** — constraint satisfaction
4. **Knapsack** — epistatic interactions
5. **No Thanks!** — co-evolutionary, no fixed landscape
6. **Checkers** — intransitive fitness $(A \succ B \succ C \succ A)$

> If composition determines diversity across **all six**, it cannot be an artifact of any particular landscape structure.



Experimental Setup

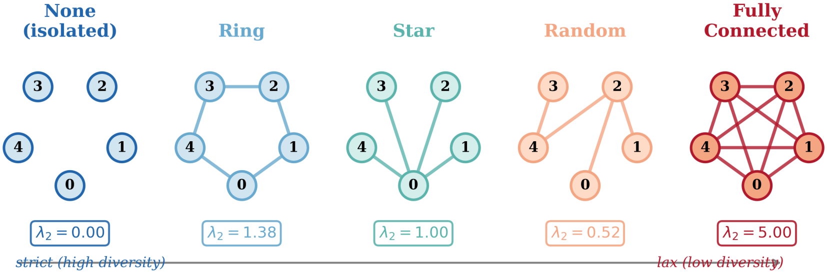
Parameter	Value
Islands	5
Individuals/island	16
Generations	100
Migration rate	0.1 every 5 gen
Topologies	5
Seeds per cell	30
Total runs	**900**

Topologies swept:

- None (isolated)
- Ring
- Star
- Random (Erdős-Rényi, resampled)
- Fully connected

Diversity metric:
Normalized pairwise Hamming (binary) / Euclidean (real-valued)

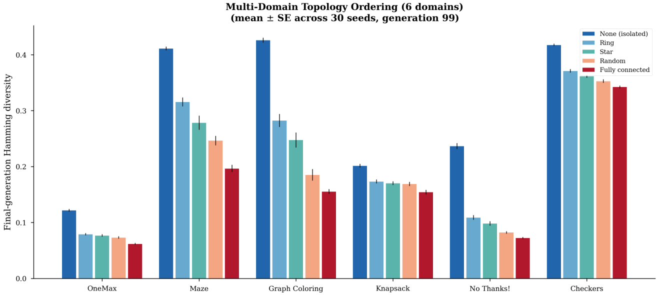
Five Topologies and Algebraic Connectivity



Key quantity: λ_2 (algebraic connectivity) — second-smallest eigenvalue of the graph Laplacian.

Smaller λ_2 ⇒ slower mixing ⇒ higher sustained diversity. none: 0 ring: 1.38 FC: 5.0

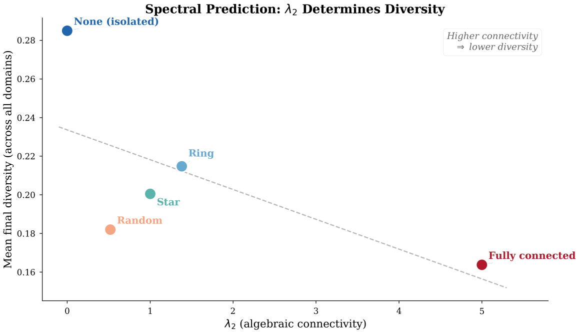
The Universal Ordering



- **6 domains.** 900 runs.
- **Identical ordering** in every domain: none > ring > star > random > FC
- **Kendall's W = 1.0** (perfect concordance)

$F_{\text{topo}} = 47.8 \quad F_{\text{domain}} = 0.13$
Domain: $p = 0.945$ (noise)
Topology > Domain: $24\times$

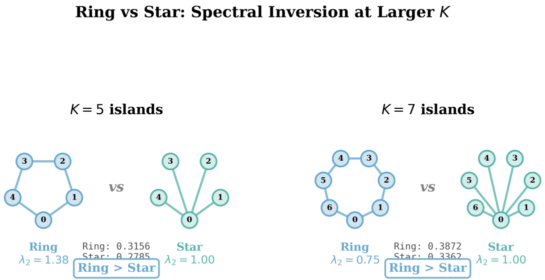
The Spectral Explanation



- Diversity follows λ_2 **monotonically**
- **None** $\rightarrow$ **Ring** transition: -35% diversity
- Each subsequent step: at most $\sim 9\%$ drop

First coupling dominates.
The symmetry-breaking step from isolation to the weakest connection loses a third of diversity.

Falsifiable Prediction: Ring/Star Inversion



	Ring	Star
$n = 5$: λ_2 (ring > star)	1.382	1.0
$n = 7$: λ_2 **(star > ring)**	0.753	1.0

Prediction: at $n = 7$, ring preserves *more* diversity than star.

Confirmed (maze, 30 seeds):
ring 0.387 ± 0.028
star 0.336 ± 0.053
$p = 6.6 \times 10^{-5}$ ✓

Scope Condition: When It Breaks

✓Holds: sufficient selective gradient

OneMax, Maze, Graph Coloring,
Knapsack, No Thanks!, Checkers

× Breaks: domain constraints dominate

Sorting Networks (7th domain)

Knowing the boundary is as valuable as knowing the result.

Four Strategy Compositions

Generational

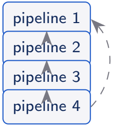
Same pipeline, repeated iteration.

Hourglass

Three explicit phases.

Island

Parallel pipelines + ring migration.

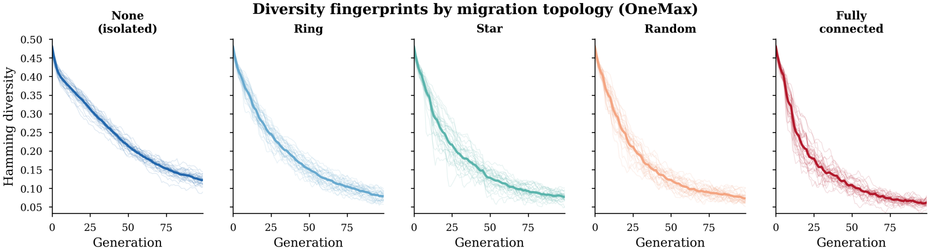


Adaptive

Plateau-triggered switch.

Same base operators (evaluate, select, crossover, mutate) — different recipes.

The Fingerprint Chart

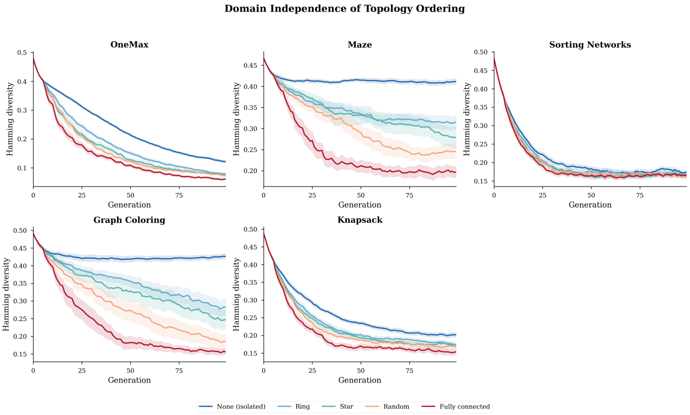


Flat: monotone decline to 0.13 **Hourglass:** crash (-85%) then rebound to 0.37

Island: diversity thermostat **Adaptive:** irreversible collapse to 0.02

Same operators. $18\times$ spread in final diversity. Shape determined by composition, not landscape.

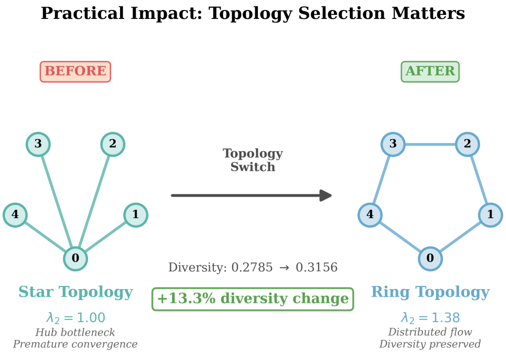
Fingerprints Are Stable Across Domains



- Hourglass crash: **same relative point** in every domain
- Island thermostat: **same relative level**
- Adaptive collapse: **equally irreversible** everywhere

Predict diversity trajectory from composition alone — before running a single experiment

Three Recommendations for Practitioners



1. **Prefer ring/mesh over star**
   Star has $\beta_1 = 0$ (no cycles).
   Add a ring: $n-1$ cycles at zero extra edge cost.
2. **Monitor λ_2, not diversity**
   Cheap (one eigenvalue).
   Predicts collapse **50–100 gen early**.
3. **Close cycles, don't raise migration rate**
   Higher rate $=$ more coupling, same topology. One edge changes both.

Summary

W = 1.0

$23.9\times$ **topology** $>$ **domain**

$18\times$ **spread in diversity**

6 domains, $p = 0.00008$

prediction $p = 6.6\times10^{-5}$

- **Composition determines diversity** — empirically, across 6 domains
- **Universal topology ordering** explained by spectral graph theory
- **Novel prediction confirmed:** ring/star inversion at $n = 7$
- **Diversity fingerprints** are properties of composition patterns, not operators
- **Categorical framework** (Kleisli arrows) makes structure visible and formally analyzable

Thank You

Composition Determines Diversity

Categorical Fingerprints of Genetic Algorithms

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Code & paper: github.com/lyra-claude/categorical-evolution

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